

COMBINATORICS FINAL EXAM

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1. (Multinomial Coefficients) A committee of 12 people is to be divided into four subcommittees of sizes 3, 4, 2, and 3.

- (a) How many ways can this be done if the four subcommittees have distinct responsibilities?
- (b) How many ways can this be done if two of the subcommittees of size 3 have identical responsibilities, while the others are distinct?
- (c) Explain why the multinomial coefficient

$$\binom{12}{3.4.2.3}$$

appears naturally in part (a).

Solution.

□

2. (Binomial Identities) Prove the identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{k}.$$

Give **two different proofs**

- (a) An algebraic proof using the binomial theorem.
- (b) A combinatorial proof by counting the same set in two ways.

Proof.

□

3. (Ramnsey Numbers)

- (a) Show that if $m \geq 3$ and $n \geq 3$ are integers then

$$R(m, n) \leq R(m-1, n) + R(m, n-1).$$

- (b) Show that if $R(m-1, n)$ and $R(m, n-1)$ are both even and greater than 2, then

$$R(m, n) \leq R(m-1, n) + R(m, n-1) - 1.$$

- (c) Show that $R(4, 3) = 9$, $R(5, 3) = 14$, and $R(4, 4) = 18$

Proof.

□

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4. (Stirling Numbers of the Second Kind) Let $S(n, k)$ denote the Stirling numbers of the second kind, the number of ways to partition an n -element set into k nonempty unlabeled blocks.

(a) Prove the recurrence

$$S(n, k) = kS(n-1, k) + S(n-1, k-1).$$

(b) Use the recurrence to compute $S(5, 2)$, $S(5, 3)$, and $S(6, 3)$.

(c) Give a direct combinatorial interpretation of $S(6, 3)$.

Solution. □

5.

Solution. □

6.

Solution. □

7.

Solution. □

8.

Solution. □

9.

Solution. □

10.

Solution. □

11.

Solution. □

12.

Solution. □

13.

Solution. □

14.

Solution. □

15.

Solution.

